

3.3.3

AI25BTECH11019 - MENAVATH SAI SANJANA

Question:

Construct an equilateral triangle $\triangle ABC$ with each side 5 cm.

Solution:

$$\mathbf{A} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 5 \\ 0 \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} x \\ y \end{pmatrix}. \quad (0.1)$$

Using the cosine rule for $\triangle ABC$:

$$BC^2 = AB^2 + AC^2 - 2(AB)(AC) \cos \angle A \quad (0.2)$$

Since $AB = AC = BC = 5$:

$$25 = 25 + 25 - 2 \cdot 25 \cos \angle A \quad (0.3)$$

From (0.3):

$$\cos \angle A = \frac{1}{2}, \quad \angle A = 60^\circ \quad (0.4)$$

Coordinates of C :

$$\mathbf{C} = \begin{pmatrix} 5 \cos 60^\circ \\ 5 \sin 60^\circ \end{pmatrix} = \begin{pmatrix} \frac{5}{2} \\ \frac{5\sqrt{3}}{2} \end{pmatrix} \quad (0.5)$$

Conclusion: The equilateral triangle $\triangle ABC$ with side 5 cm has vertices

$$\mathbf{A} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 5 \\ 0 \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} \frac{5}{2} \\ \frac{5\sqrt{3}}{2} \end{pmatrix}. \quad (0.6)$$

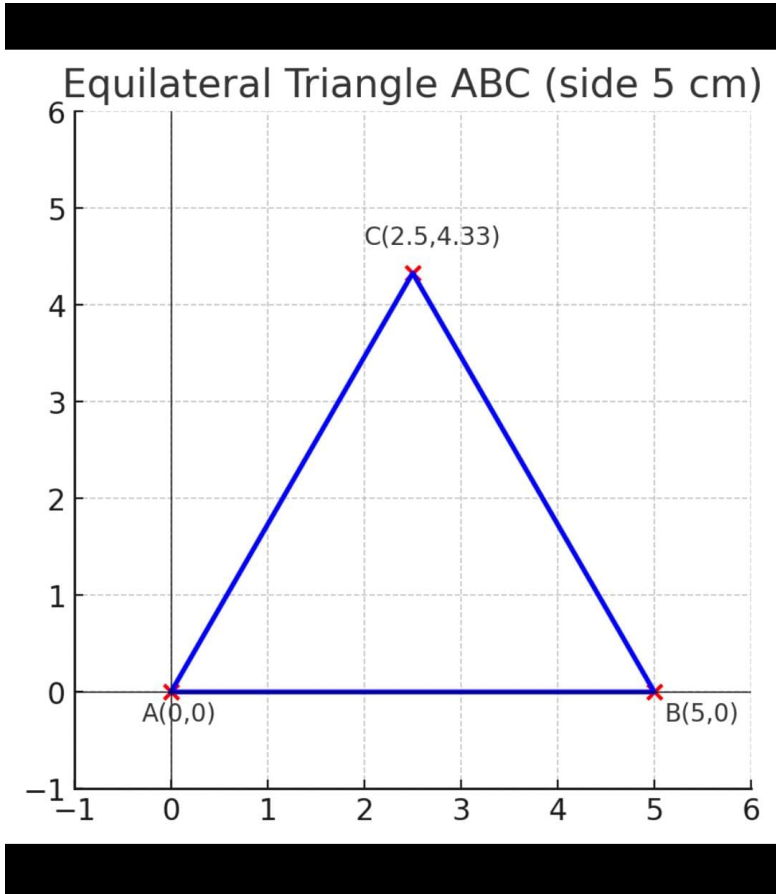


Fig. 0.1